THE NOVA RESEARCH CURRICULUM

Mathematics

Grade 5 · Term 1

Student edition

Outcomes: MA-5.1 numbers to 10 million (Pakistani and international systems), operations and order of operations; MA-5.2 HCF and LCM.

Lesson notes

MA-5.1 Large numbers and operations (Weeks 1–8)

International system: thousands, millions (7,250,000). Pakistani system: lakh and crore (72,50,000 = 72 lakh 50 thousand; 1 crore = 10 million). Convert between them. All four operations with large numbers; order of operations with brackets (brackets first, then × and ÷, then + and −).

MA-5.2 HCF and LCM (Weeks 9–12)

Prime factorisation with factor trees and the division method. HCF = the largest number that divides all; LCM = the smallest number all divide into. Real problems: cutting ribbons into the longest equal pieces (HCF); when two buses next leave together (LCM).

Workbook pages

Page	Outcome	Activity
1–2	MA-5.1	Place-value charts in both systems; conversions.
3–4	MA-5.1	Operations with large numbers; brackets puzzles.
5–7	MA-5.2	Factor trees; HCF and LCM by division; word problems.

Question bank

#	Outcome	Question	Your answer
1	MA-5.1	1 crore =	1 million / 10 million / 100,000 / 100 million
2	MA-5.1	Write 7,250,000 in the Pakistani system.	_____
3	MA-5.1	25 lakh =	250,000 / 2,500,000 / 25,000,000 / 25,000
4	MA-5.1	8 + 4 × 3 =	36 / 20 / 15 / 24
5	MA-5.1	(8 + 4) × 3 =	20 / 36 / 15 / 24
6	MA-5.1	45 ÷ (5 + 4) =	13 / 9 / 5 / 4
7	MA-5.1	Brackets are worked out first. True or false.	True / False
8	MA-5.2	HCF of 12 and 18 =	3 / 6 / 12 / 36

#	Outcome	Question	Your answer
9	MA-5.2	LCM of 4 and 6 =	2 / 10 / 12 / 24
10	MA-5.2	HCF of 24, 36 and 60 =	6 / 12 / 24 / 4
11	MA-5.2	LCM of 6, 8 and 12 =	12 / 24 / 48 / 96
12	MA-5.2	Two buses leave every 12 and 18 minutes. After how many minutes do they leave together again?	_____
13	MA-5.2	Ribbons of 24 cm and 36 cm are cut into equal pieces as long as possible. How long is each piece?	_____